

# Varun Nayak

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Product-oriented **Robotics Software Engineer** with 6+ years of experience building deployment-ready robotics applications.

## EXPERIENCE

**INTRINSIC** at Google, Mountain View, CA

**Senior Robotics Software Engineer** July 2026 - Present

- Currently working on accelerating deployments across North America.

**DEXTERITY.AI** Warehouse Robotics, Redwood City, CA

**Senior Robotics Engineer** July 2022 – July 2026

- **Architected and developed Dexterity's robot applications**, including orchestration, task allocation, sensor integration, configuration management, ML model integration, and debugging and visualization tools.
- **Architected Dexterity's robot developer platform** using REST APIs, enabling partners and solution engineers to build and integrate custom robotic automation workflows for faster deployment.
- **Built end-to-end simulation infrastructure** using PyBullet, Drake, and in-house physics engines for automated testing and validation across diverse warehouse environments.
- **Re-designed the application stack into an event-driven framework** using Python asyncio, enabling a patented multi-robot coordination algorithm that improved system performance by 40%.

**Robotics Engineer** July 2020 - July 2022

- **Developed robust robot control systems** using operational space control framework, trajectory optimization, and sampling-based path planning (Python/C++), establishing dexterity's core robotics platform.
- **Successfully deployed and maintained 100+ robots** in production environments for kitting, palletizing, depalletizing and trucking loading operations, demonstrating real-world impact and system reliability.
- **Implemented comprehensive state management, error handling, and metrics collection** systems, ensuring robust operation of robotic applications in production environments.

**AURIS HEALTH, INC.** Surgical Robotics, Redwood City, CA

Robotics Software Intern | Summer 2019

- **Implemented and deployed impedance-based control** for multi-DoF robot arms in C++, and clinical engineering UIs (using QT) for tuning them.
- **Reduced surgeon preparation time by 20%** through improved control while complying with FDA regulations for robotic safety.

## SELECTED ACADEMIC PROJECTS

**Autonomous Navigation Stack:** Built complete navigation infrastructure using ROS, implementing A\* path planning, EKF-based SLAM, using LiDAR perception.

**Robotic Arms:** Developed comprehensive perception-to-control pipeline for 7-DoF robotic arms, including forward/inverse kinematics and trajectory generation.

**Reinforcement Learning Velocity Planner:** Implemented Q-Learning and SARSA reinforcement algorithms to train velocity planners in simulation.

## TEACHING/VOLUNTEERING

**Teaching Assistant:** Introduction to Robotics with Dr. Oussama Khatib, Stanford University (2020)

**Teaching Assistant:** Dynamic Systems, Vibration & Control, Stanford University (2019)

**Robotics Workshop Instructor, cybermath.org** (2019): Taught robotics fundamentals to middle school students.

## EDUCATION

**STANFORD UNIVERSITY**

MASTER OF SCIENCE, ROBOTICS (2020)

**Coursework:** Machine Learning, Computer Vision, Robot Control, Motion Planning, Optimization, Mechatronics.

**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI, INDIA**

BACHELOR OF SCIENCE, MECHANICAL ENGINEERING (2018)

## SKILLS

**PROGRAMMING**

Python • C/C++ • Golang

**LANGUAGES**

**ML/AI & ROBOTICS**

PyTorch • ROS • Isaac Sim • Drake • PyBullet • numpy • scipy

**INFRASTRUCTURE & TOOLS**

Google Cloud • Kubernetes • Docker • GitLab CI • REST • gRPC • asyncio • FastAPI

**MONITORING & DATA**

Prometheus/Grafana • Jaeger • Elasticsearch/Kibana • Redis • SQL/BigQuery

## PATENTS/PUBLICATIONS

**Patent (System & Software):**

Multiple robot simultaneous and synchronous pick and place

**Patent (System & Software):** Tray handling autonomous robot

**Patent (Design):** Multi-mode robotic end effector

**Thesis:** Design and control of an aerial manipulator for contact-based inspection